

DEEP NEMA

Banswara, Rajasthan

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Professional Summary

Generative AI and Machine Learning enthusiast with hands-on experience developing RAG applications, Computer Vision systems, and real-time analytics platforms. Skilled in Python, SQL, Machine Learning, REST APIs, and scalable software development with a strong foundation in problem solving, debugging, and intelligent system design.

Education

Vellore Institute of Technology

2022 – Present

B.Tech in Computer Science (Health Informatics)

Technical Skills

Programming: Python, SQL, Java

AI/ML: Machine Learning, Generative AI, RAG, Computer Vision, Data Preprocessing, Model Evaluation

Technologies: FastAPI, REST APIs, Streamlit, JSON, MySQL

Tools: Git, GitHub, Linux Basics, MS Office

Core Concepts: DSA, OOP, DBMS, Problem Solving, Debugging, Troubleshooting

Experience

AI/ML Computational Science Intern – Saiket Systems

Oct 2025 – Apr 2026

Converser Tool (GenAI NLP Platform)

- Developed a GenAI-powered chatbot using Google's Gemini model, achieving 85 percent accuracy in generating optimized BigQuery scripts from natural language prompts and improving query generation efficiency by 80 percent.
- Designed and implemented prompt engineering, NLP, embeddings-based semantic search, and rule-based logic to generate accurate database queries, improving query precision by 30 percent across diverse question structures.
- Collaborated with leadership, business analysts, and developers to refine product logic and enhance chatbot usability, delivering human-readable responses and improving user experience and interpretability by 30 percent.
- Implemented Role Level Security (RLS) and optimized application performance, reducing response time by 60 percent, improving system accuracy by 30 percent, and strengthening secure data accessibility.

Skills: Gemini LLM || Generative AI || Prompt Engineering || NLP || Python || SQL || BigQuery || GCP || Embeddings || Semantic Search || Conversational AI || RAG || Machine Learning || Role Level Security (RLS)

Projects

Enterprise Knowledge Assistant (RAG-Based LLM System)

GitHub

- Developed a Retrieval-Augmented Generation platform using LLMs, FAISS, and vector search for intelligent document question answering.
- Implemented document ingestion, embedding generation, semantic retrieval, and response generation pipelines.
- Built scalable workflows for processing and querying large document repositories.

Intelligent Video Analytics Platform

GitHub

- Developed a real-time video intelligence system using Computer Vision and Deep Learning techniques.
- Built low-latency inference APIs using FastAPI and OpenCV for scalable video processing.
- Implemented automated object detection and scene analysis pipelines for monitoring applications.

Real-Time Financial Intelligence Dashboard

GitHub

- Developed a real-time analytics platform processing 1000+ cryptocurrency records daily through API-driven pipelines.
- Implemented automated data collection, transformation, and visualization workflows.
- Built interactive dashboards enabling market trend monitoring and exploratory data analysis.

Certifications

- Oracle Certified Professional – Java SE 11 Developer
- Generative AI using IBM Watsonx – IBM
- Applied Machine Learning in Python – Coursera